Ocean Engineering 357 (2026) 125460



Contents lists available at ScienceDirect

Ocean Engineering

journal homepage: www.elsevier.com/locate/oceaneng



Research paper

Experimental and NDEM investigation of resistance and pressure statistics of an inclined plate interacting with broken ice

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ARTICLE INFO

Keywords:
Plate–ice interaction
Experiment
NDEM
Ice resistance and pressure
Probability statistics

ABSTRACT

Wide hull panels operating in broken ice experience intermittent multi-body impacts, transient jamming, and highly non-uniform pressures, which cannot be characterized by mean resistance alone. This paper investigates an inclined plate advancing through floating broken ice using towing-tank experiments and an impulse-based non-smooth discrete element method (NDEM). Global towing resistance is measured by a force bracket, and sensel-scale normal pressures are captured by a Tactilus array. Collision-pressure statistics are extracted by referencing the pressure record to an open-water baseline and trimming background fluctuations. To enable sensor-consistent local pressure monitoring in an impulse-based solver, the plate face is represented as a rigid high-inertia cell assembly matching the 20 × 8 sensel footprint, and cell-wise ice loads are identified via an impulse-momentum inversion from discrete state increments. A conservative impulse-to-cell reconstruction then yields step-averaged normal pressures consistent with the tactile measurements. The NDEM predictions agree well with experiments in resistance level and in the burst-like, heavy-tailed collision-pressure statistics. Parametric results show that increasing inclination angle and ice concentration mainly increase the occurrence and persistence of strong-contact states through intermittent frontal accumulation and force-chain-like load transfer, thereby extending the heavy tails of both resistance and collision pressure, whereas increasing towing speed mainly scales the intensity of extreme events. The proposed framework enables efficient prediction of both global resistance and local extreme-load statistics in broken-ice operations.

1. Introduction

The rapid reduction of polar sea-ice extent and the associated expansion of marginal ice zones have reshaped the operational envelope of Arctic navigation (Wang et al., 2020). With improved seasonal accessibility, commercial and research vessels increasingly encounter ice environments dominated not by level ice, but by fragmented floes and broken-ice fields spanning low-to-medium concentrations (Ng et al., 2018; Stephenson et al., 2018). Under such conditions, ship-ice interaction is governed by frequent multi-body collisions, intermittent squeezing, and sustained sliding in icebreaking channels or naturally fragmented pack ice, where floes drift, rotate, locally raft, and accumulate along the hull under the combined action of contact and hydrodynamic forces. Although individual small floes seldom cause catastrophic structural damage, their collective effect can markedly increase resistance, degrade maneuverability, and contribute to ice trapping in severe cases. This motivates reliable prediction methods that can address both global resistance and local load extremes in broken-ice operations (Huang et al., 2021; Sazonov and Dobrodeev, 2021).

From an engineering standpoint, many critical polar scenarios involve wide contact surfaces interacting with broken ice at nontrivial attitudes (Chen et al., 2024). Typical examples include shoulder and waterline regions of a polar ship operating in broken-ice channels, whose wide hull panels experience ice impact during course-keeping and maneuvering (Zhong et al., 2023). In these cases, the interaction cannot be adequately represented by a single resultant force alone: the spatial and temporal variability of local pressure governs structural design margins, fatigue accumulation, and extreme-event assessment. Accordingly, design rules often idealize the design-driving ice-load case using an equivalent uniform patch pressure; while convenient for engineering practice, this idealization does not resolve localized, short-duration pressure maxima (IACS, 2016). Field measurements and

This article is part of a special issue entitled: OE_PolarShip published in Ocean Engineering.

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https://doi.org/10.1016/j.oceaneng.2026.125460
Received 17 January 2026; Received in revised form 5 March 2026; Accepted 4 April 2026
Available online 10 April 2026
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laboratory tests have repeatedly shown that ice loads are strongly non-uniform in space and highly intermittent in time, including pronounced peak events associated with transient contact states and load concentrations (Riska et al., 1983; Masterson and Frederking, 1993; Daley et al., 2007; Ritch et al., 2008; Quinton, 2015). These observations collectively imply that a methodology capable of jointly quantifying resistance and high-frequency, non-uniform pressure in broken ice is essential, yet remains challenging because the loading process is random, intermittent, and coupled with ice motion and water effects (Sun and Huang, 2021).

Model-scale experiments and field measurements remain indispensable for revealing the underlying physics. Full-scale monitoring has traditionally inferred ice-induced loads from strains measured on internal frames and then converted them to external loads; uncertainties arise from structural transfer functions, limited sensor coverage, and the coexistence of wave/hydrodynamic loads with ice-induced components (Suominen et al., 2017; Jeon et al., 2018). Statistical characterization of peaks is commonly performed using probability distributions (e.g., Weibull or exponential families), and separation techniques such as Rayleigh-based methods or data-driven regressors have been explored to extract ice-induced components from mixed signals (Suominen and Kujala, 2014; Shamaei et al., 2020; Wu et al., 2021). At model scale, tactile sensors provide a direct route to pressure measurement and have been used in level-ice or controlled impact contexts (Kujala and Arughadhoss, 2012), while continuous flexible tactile sensing has shown promise in capturing spatially varying impact pressure during glancing interactions (Sun and Huang, 2020, 2021). For broken-ice fields, however, the randomness of floe contacts makes discrete sensor placement prone to missing critical events, and even with dense arrays, interpreting pressure signals requires robust processing to separate hydrostatic/hydrodynamic contributions from collision-induced pressures and to quantify probability characteristics in a consistent manner across experiments and simulations.

In parallel, numerical simulation has become an increasingly important complement to experiments due to controllability, cost efficiency, and the ability to resolve multi-body kinematics (Ni et al., 2020). A broad spectrum of approaches-Discrete Element Method (DEM), Finite Element Method (FEM), Peridynamics (PD)-has been applied to ship-ice and ice-structure interaction (Lau et al., 2008; Kim et al., 2013; Xue et al., 2021; Huang et al., 2021). Among them, DEM-type approaches are particularly suitable for fragmented ice due to their intrinsic discrete representation, evolving from early 2D disk-element models (Hansen and Løset, 1999) to 3D polyhedral formulations and coupled hydrodynamics treatments (Liu and Ji, 2018; Xue et al., 2024). Nevertheless, high-fidelity coupling remains computationally demanding when thousands to tens of thousands of bodies and high-frequency contacts are involved, limiting routine use for broad parametric studies.

A computationally efficient alternative is the non-smooth discrete element method (NDEM), also referred to as contact dynamics or impulse/constraint-based rigid-body dynamics (Konno and Mizuki, 2006; Yang et al., 2021). In contrast to smooth DEM (penalty contact laws with explicit integration), NDEM formulates multi-body contact through complementarity/constraint problems with implicit or semi-implicit stepping and iterative solvers, enabling stable simulation with larger time steps and high collision throughput (Metrikin et al., 2012; Yulmetov et al., 2016; Van den Berg et al., 2018). Practical implementations in open-source rigid-body engines (e.g., Bullet, ODE, Box2D) provide efficient collision detection pipelines and constraint solvers, and have been adopted in ice engineering studies where collision-dominated dynamics and computational efficiency are central (Catto, 2006; Tuhkuri and Polojärvi, 2018). Within ship-ice applications, NDEM has progressed from resistance prediction and mechanism analysis in broken ice to more operation-oriented problems such as channel/boundary effects, maneuvering in broken ice areas, and maneuverability modelling in level ice (Lubbad and Løset, 2011; Janβen et al., 2017; Yang et al., 2020, 2024). Meanwhile, data-driven/meta-modelling strategies have been introduced to accelerate resistance estimation within NDEM frameworks (Sun et al., 2023). Despite this progress, two methodological issues continue to constrain NDEM when local pressure is a primary target: (i) contact forces are not always directly accessible or physically smooth in impulse-based solvers, and step-wise forces inferred from impulses can fluctuate strongly in dense collision systems; and (ii) converting intermittent multi-point contact information into a measurement-consistent pressure field comparable to tactile sensor arrays remains nontrivial when contacts appear/disappear rapidly and contact multiplicity varies over time.

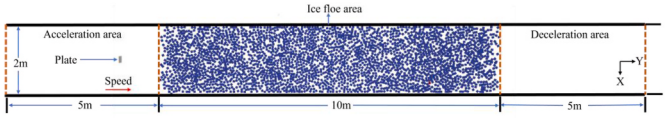


Fig. 1. Schematic of the towing tank and the three sections used in the experiments.

The literature still lacks directly comparable experiment–NDEM studies that resolve resistance evolution together with the spatiotemporal statistics of collision-induced pressure in broken-ice interaction. Recent reviews on broken-ice simulations emphasize that most existing computational models primarily target resistance estimation and encourage more attention to local ice-load estimation, while ice model tests increasingly pair resistance measurements with tactile pressure arrays to provide distributed local pressure information. Despite this experimental capability, a directly comparable, sensor-consistent local pressure output is rarely available in impulse-based non-smooth simulations, because an impulse-based time-stepping formulation is built on impulse–momentum updates and does not explicitly resolve impulsive forces, and because contact topology in broken ice evolves rapidly in space and time. Moreover, a simplified building-block structure is valuable for isolating mechanisms and developing transferable post-processing methods beyond a single ship geometry. In this study, we conduct towing-tank experiments of an inclined/vertical flat plate interacting with floating broken ice floes and develop a validated NDEM counterpart. Two measurement-consistent procedures are introduced: an impulse–momentum reconstruction to infer resistance from discrete kinematics, and a cell-resolved conservative reconstruction to obtain pressure fields on a sensor-like 20 × 8 grid. Section 2 describes the experimental setup and sensing strategy; Section 3 presents the numerical framework and reconstruction methods; Section 4 reports experimental–numerical comparisons and parametric effects; and Section 5 summarizes the main findings and implications.

2. Experimental setup

Model tests were conducted in the outdoor towing tank at Harbin Engineering University (HEU). The tank is 20 m long, 2 m wide, and 1.2 m deep, with a still-water depth of approximately 0.9 m. The towing track was divided into three consecutive sections (Fig. 1): a 5 m open-water acceleration region, a 10 m broken-ice test section, and a 5 m open-water deceleration region. The carriage speed is continuously controllable (0-2.0 m/s). Unless otherwise stated, all force and pressure analyses use the quasi-steady segment during the plate passage through the 10 m broken-ice section. A right-handed coordinate system is defined with Y along the towing direction, X transverse, and Z upward.

To represent a canonical “wide surface” interacting with broken ice, analogous to a local broad hull patch near the shoulder/waterline region of a polar ship (Fig. 2), the test body was idealized as a rigid flat plate. In the prototype, a representative panel around the waterline with an approximate width of B = 6.4 m and a height of H = 8.0 m was selected, with a corresponding draft of about 4.8 m. Considering the geometric constraints of the towing tank, a geometric scale ratio of $\lambda = 32$ was adopted, yielding a model plate with a width of $B_m = 0.20$ m and a

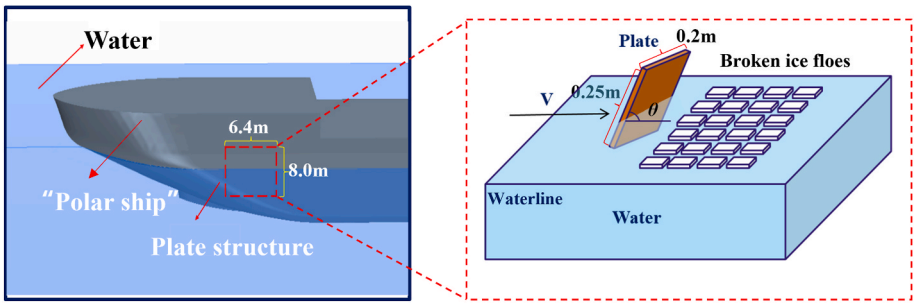


Fig. 2. Schematic of the plate structure and the corresponding plate-broken ice floes interaction scenario.



(a) $\eta = 40\%$ (b) $\eta = 60\%$ (c) $\eta = 80\%$

Fig. 3. Bird's eye view of the initial test conditions for different ice concentrations.

height of $H_m = 0.25$ m. A fixed waterline condition was enforced in all tests: the submerged height of the plate was maintained at 0.15 m, while the emerged height was 0.10 m. This design isolates the effects of inclination and ice concentration by keeping the hydrostatic reference level unchanged when the plate attitude varies.

The speed of the plate is determined by the Froude similarity criterion:

$$Fr = \frac{V}{\sqrt{gL}} \tag{1}$$

where V and L represent the plate speed and overall width of plate, respectively and g is gravitational acceleration. The specific speeds in the experiments are taken as 0.5, 0.59 and 0.68 m/s, corresponding to prototype speeds of 5.5, 6.5 and 7.5 kn ($\lambda = 32$).

Non-refrigerated synthetic model ice made of polypropylene (PP) was used following established practice for broken-ice towing tests (Zong et al., 2020). The floes were square plates with an average side length of about 0.067 m and a thickness of about 0.015 m, corresponding to a prototype thickness of approximately 0.48 m and a characteristic floe size of about 2.1 m. Such a simplified geometry is widely used to ensure a controlled and repeatable broken-ice field in towing-tank experiments (Liu, 2018). The present study targets a collision-sliding-accumulation dominated regime; therefore, secondary fragmentation of the floes during towing was not considered. To ensure that the synthetic pieces remain intact, the floe dimension was selected to be equal to or smaller than the critical length l proposed by Lu et al. (2016):

$$l = \sqrt[4]{\frac{K}{\rho_w g}} \tag{2}$$

where ρ_w is the water density. K is the flexural rigidity of the sea ice

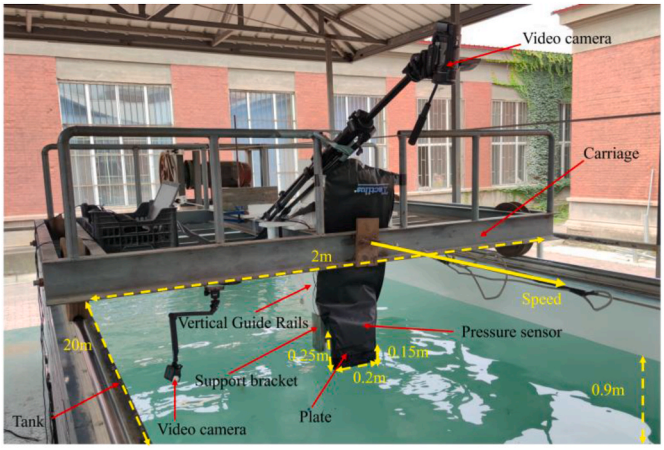


Fig. 4. Photograph of the towing-tank experimental setup for plate-broken ice interaction.

given by

$$K = \frac{EH^3}{12(1-\nu^2)} \tag{3}$$

in which E and ν are Young's modulus and Poisson ratio of sea ice, respectively, $E = 1.0$ GPa and $\nu = 0.3$, and H is the thickness of the broken ice. Accordingly, considering Froude and Cauchy similarity criteria (Huang et al., 2018), l is defined as 0.1761 m as per Eq. (2).

Ice concentration η is defined as the areal coverage of floating pieces within the test section. According to the relevant survey of Arctic ice conditions (Wang, 2019; Li, 2019), three concentrations were considered, namely $\eta = 40\%$, 60% and 80%, as illustrated by the bird's eye

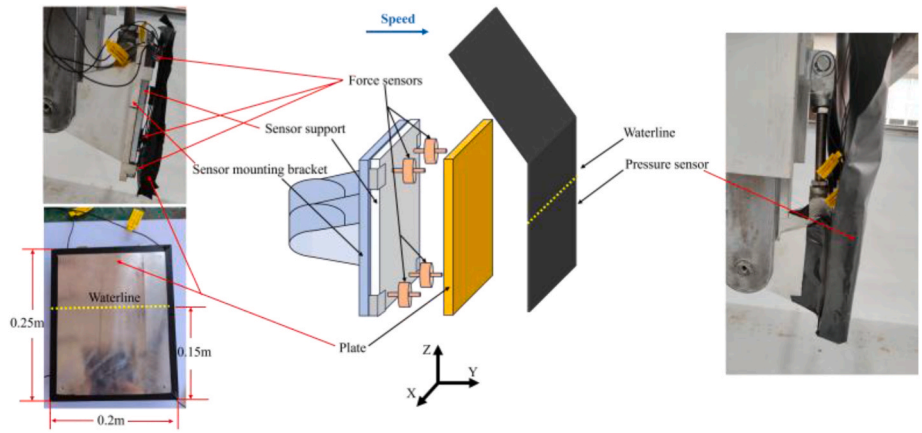


Fig. 5. Arrangement of the resistance sensors and the tactilus sensel on the plate.

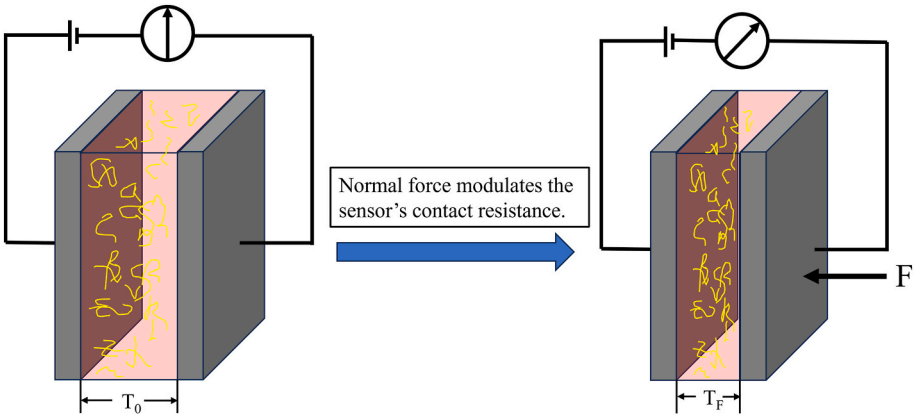


Fig. 6. Working principle of a piezoresistive tactile sensor (Wan et al., 2017).

views of the initial conditions (Fig. 3). For each run, the required number of floes was randomly placed and manually redistributed to obtain an approximately uniform initial field, and the ice was allowed to settle to a quasi-static state before towing.

The overall experimental arrangement is illustrated in Fig. 4. The flat plate was mounted beneath the towing carriage through a rigid support frame, where vertical guide rails were used to maintain the prescribed heave position and to suppress lateral drift during towing. A piezoresistive tactilus sensor was attached to the plate face, and synchronized video cameras were deployed on the carriage and on a fixed bracket to record the plate-ice interaction from multiple viewpoints for subsequent interpretation of collision, sliding, and local accumulation processes.

Two types of measurements were acquired: (i) global towing-direction resistance and (ii) spatially resolved normal pressure on the plate. The global resistance was measured by four force sensors integrated into the mounting bracket (rated capacity 100 N per sensor; combined error ±0.03%), and the towing-direction component of the summed signal was taken as the instantaneous total resistance (Fig. 5).

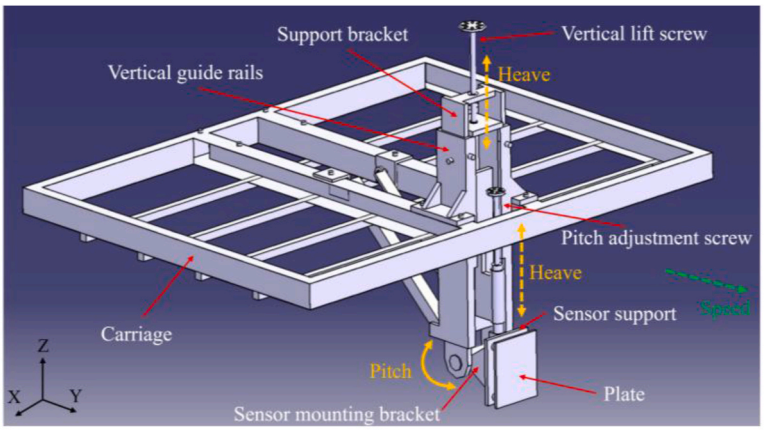
The local pressure was measured using a piezoresistive tactilus sensor attached to the plate surface (Fig. 5). The employed tactilus sensor comprises 640 sensing elements arranged as 32 columns along the sensor length and 20 rows along its width. Owing to the limited plate area and the prescribed mounting layout, only a sub-array located at the sensor head was utilized in the present tests; specifically, a 20 (width direction) × 8 (height direction) region was bonded to the plate surface, providing 160 individual sensing units ("sensels") uniformly distributed over the instrumented area (Fig. 5). Each sensel covers approximately 10.0 mm (width) × 33.1 mm (height), consistent with tactilus sensor specifications commonly adopted for hull pressure measurements in ice model tests (Kujala and Arughadhoss, 2012; Sun and Huang, 2020; Zhong et al., 2023). The sensor measures normal pressure only (no tangential shear), which is appropriate for the present focus on impact/crushing-type pressure peaks and their statistics. The nominal measurement range of the tactilus sensor is up to 345 kPa. Pressure frames were recorded at 50 Hz. The working principle is illustrated in Fig. 6: the applied normal load modulates the contact resistance inside the sensor, and the measured electrical signal is converted to pressure through calibration (Wan et al., 2017). In post-processing, pressure fields can be partitioned relative to the fixed waterline (above-water, waterline-adjacent, and submerged bands) for consistent statistical comparisons.

The plate inclination angle was set using a pitch adjustment mechanism shown in Fig. 7 to $\theta = 30°$, 45°, 60°, and 90° (vertical plate). For each inclination, the heave position was re-adjusted to keep the waterline fixed on the plate surface before towing. For each condition (η, θ, V), three repeated runs were performed; after each run, the floes were re-randomized and the field was re-prepared to reduce memory effects, and the reported statistics were obtained from the ensemble of repeated tests.

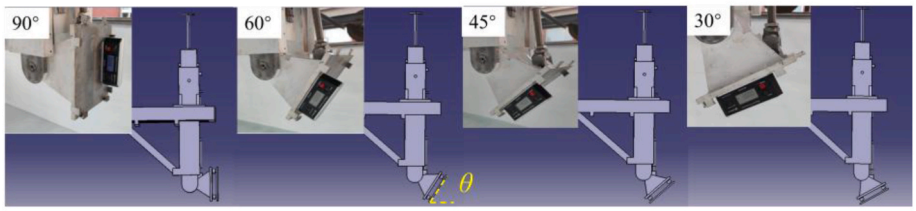
3. Numerical methods and setup

3.1. Collision detection

The simulations were performed using a Bullet-based non-smooth discrete element method (NDEM) framework. Impulse-based NDEM/contact-dynamics formulations have been widely used for ice–structure and ship–ice interaction problems, particularly in broken-ice fields



(a) Overall assembly



(b) Inclination-angle settings

Fig. 7. Plate mounting and attitude-control system.

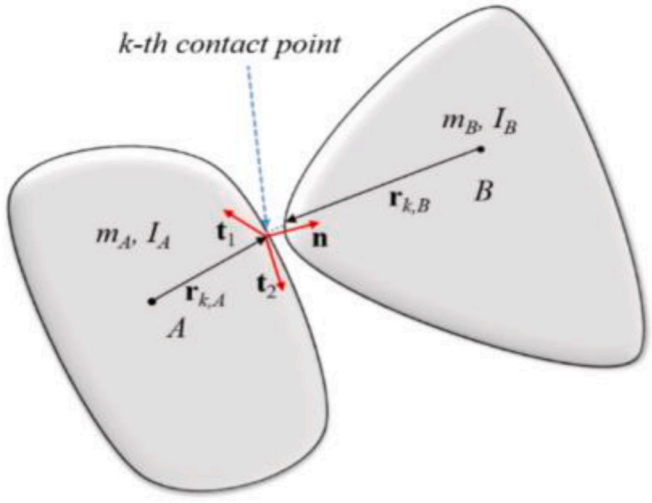


Fig. 8. Contact k between bodies A and B (Yang et al., 2025).

where dense multi-body contacts and frequent contact switching are dominant, and physics-engine-based implementations have also been examined for broken-ice dynamics in the literature (Lubbad and Løset, 2011; Janβen et al., 2017). In the present study, Bullet's collision-detection and impulse/constraint-solving kernels are adopted as established building blocks (without modifying the underlying algorithms), while the methodological contributions are introduced later in the load identification procedures. Contact pairs are identified using Bullet's standard two-stage pipeline: a broadphase Sweep-and-Prune (SAP) scheme based on axis-aligned bounding boxes (AABBs), followed by a narrowphase Gilbert-Johnson-Keerthi (GJK) procedure to determine contact existence and contact kinematics (Yang et al., 2020). This configuration enables efficient handling of the dense multi-body interactions typical of broken-ice fields and provides stable contact information for subsequent impulse/constraint resolution in NDEM-based ship/structure–ice simulations (Yang et al., 2024).

3.2. Constraint computation

Contacts are resolved within an impulse/constraint formulation by using Lagrange multipliers, enforcing unilateral non-penetration and Coulomb friction via the Signorini-Coulomb law, while energy dissipation is represented by restitution coefficients. For a contact point k between bodies A and B (Fig. 8), the discrete momentum balance over one time step Δt is written as Eqs. (4) and (5), where the contact impulse $\boldsymbol{\lambda}_{\mathrm{cont},k}$ is decomposed into normal and tangential components as Eqs. (6) and (7). The normal complementarity condition and the friction bounds follow Eq. (8), with the static and kinetic friction coefficients taken equal in the present study.

The momentum equation of bodies A and B in the collision system can be written as:

$$m_i \Delta \boldsymbol{v}_i = \boldsymbol{F}_{\mathrm{ext},i} \Delta t + \sum_k \boldsymbol{\lambda}_{\mathrm{cont},k} \tag{4}$$

$$\boldsymbol{I}_i \Delta \boldsymbol{\Omega}_i = \boldsymbol{\tau}_{\mathrm{ext},i} \Delta t + \sum_k \boldsymbol{r}_{k,i} \times \boldsymbol{\lambda}_{\mathrm{cont},k} \tag{5}$$

where m_i and I_i represent the mass and moment of inertia of the body i, respectively; $\Delta \boldsymbol{v}_i$ and $\Delta \boldsymbol{\Omega}_i$ denote the changes in the body's velocity and angular velocity over the time step Δt; $\boldsymbol{F}_{\mathrm{ext},i}$ and $\boldsymbol{\tau}_{\mathrm{ext},i}$ are the external force and torque excluding the collision force; $\boldsymbol{r}_{k,i}$ is the position vector from the center of mass of the object to the collision point k; and $\boldsymbol{\lambda}_{\mathrm{cont},k}$ is the collision impulse at collision point k, which can be derived as follows:

$$\lambda^n_{\mathrm{cont},k} = \frac{(\Delta \boldsymbol{v}_{\mathrm{rel}} \cdot \boldsymbol{n}) \cdot \boldsymbol{n}}{1/m_A + 1/m_B + (\boldsymbol{r}_{k,A} \times \boldsymbol{n}/\boldsymbol{I}_A) \times \boldsymbol{r}_{k,A} \cdot \boldsymbol{n} + (\boldsymbol{r}_{k,B} \times \boldsymbol{n}/\boldsymbol{I}_B) \times \boldsymbol{r}_{k,B} \cdot \boldsymbol{n})} \tag{6}$$

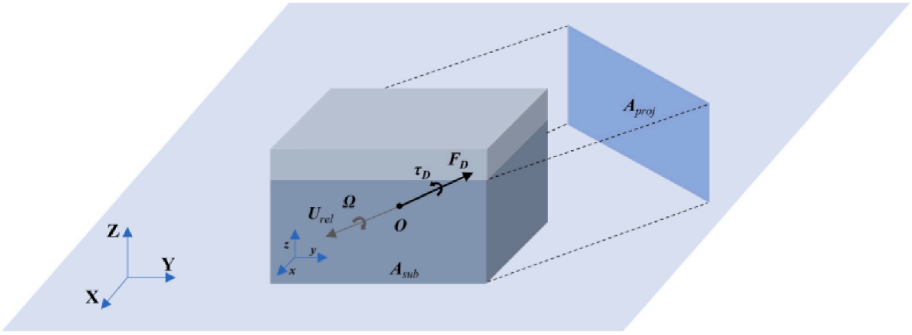


Fig. 9. Hydrodynamic force acting on the ice floe.

$$\lambda^t_{\text{cont},k} = \frac{(\Delta \boldsymbol{v}_{\text{rel}} \cdot \boldsymbol{t}) \cdot \boldsymbol{t}}{1/m_A + 1/m_B + (\boldsymbol{r}_{k,A} \times \boldsymbol{t}/\boldsymbol{I}_A) \times \boldsymbol{r}_{k,A} \cdot \boldsymbol{t} + (\boldsymbol{r}_{k,B} \times \boldsymbol{t}/\boldsymbol{I}_B) \times \boldsymbol{r}_{k,B} \cdot \boldsymbol{t})} \tag{7}$$

where $\lambda^n_{\text{cont},k}$ and $\lambda^t_{\text{cont},k}$ represent the normal and tangential components of the contact impulse, respectively; $\Delta \boldsymbol{v}_{\text{rel}} = \boldsymbol{v}_{\text{rel}}{}^{+} - \boldsymbol{v}_{\text{rel}}{}^{-}$ is the change in relative velocity at the contact point. The coefficient of restitution e is introduced to describe the relationship between velocities before and after the collision: $\boldsymbol{v}_{\text{rel}}{}^{+} = -\,\text{e} \cdot \boldsymbol{v}_{\text{rel}}{}^{-}$.

The normal velocity and the normal contact impulse constitute the complementarity condition that if one of them is positive and the other one must be zero. Meanwhile, the tangential impulse is limited by the Coulomb's friction law:

$$-\mu \left\| \lambda^n_{\text{cont},k} \right\| \le \lambda^t_{\text{cont},k} \le \mu \left\| \lambda^n_{\text{cont},k} \right\| \tag{8}$$

where μ is the coefficient of friction.

3.3. Time integration

The time stepping follows the standard semi-implicit Euler update used in impulse-based rigid-body solvers: velocities are updated by external and contact/constraint impulses, and positions and orientations are then integrated using the updated velocities:

$$\boldsymbol{v}_i^{t+\Delta t} = \boldsymbol{v}_i^t + \Delta \boldsymbol{v}_i \tag{9}$$

$$\boldsymbol{\Omega}_i^{t+\Delta t} = \boldsymbol{\Omega}_i^t + \Delta \boldsymbol{\Omega}_i \tag{10}$$

$$\boldsymbol{x}_i^{t+\Delta t} = \boldsymbol{x}_i^t + \boldsymbol{v}_i^{t+\Delta t}\, \Delta t \tag{11}$$

$$\boldsymbol{\psi}_i^{t+\Delta t} = \boldsymbol{\psi}_i^t + \boldsymbol{\Omega}_i^{t+\Delta t}\, \Delta t \tag{12}$$

where x_i and ψ_i represent the position and rotation of body i.

The contact impulse was converted to an equivalent contact force according to Yulmetov et al. (2016), and the collision duration Δt should be not larger than:

$$\Delta t \approx \frac{\pi}{2} \sqrt{\frac{M^*}{K}} \tag{13}$$

where M^* is the reduced mass and K is the contact stiffness, in which the reduced mass is equal to the ice floe's mass in the collision in the air. By contrast, in the collisions in the water, the hydrodynamic added mass can significantly modify the floe inertia, and the near-hull added mass may exceed the far-field value (Zong et al., 2019). Following Yang et al. (2021), an added-mass coefficient $\alpha = 2.0$ is adopted, such that $M^* \approx (1+\alpha) M_{ice}$. Together with $K = 1.0 \times 10^6$ N/m, Eq. (13) gives a collision duration of $\Delta t = 0.000676$ s. Therefore, the simulation time step is chosen as 0.0005 s in this paper.

3.4. Fluid force model

Hydrodynamic forces acting on an individual floe are modeled as buoyancy plus drag. Buoyancy forces and moments are obtained by hydrostatic integration over triangulated submerged surfaces (Catto, 2006). The drag force $\boldsymbol{F}_D$ and drag moment $\boldsymbol{\tau}_{\text{D}}$ are evaluated using Morison's equation (Fig. 9; Eqs. (14) and (15)), following common practice in NDEM-based ice–water simulations (Yang et al., 2021).

$$\boldsymbol{F}_D = -0.5 C_D \rho_w A_{\text{proj}} \boldsymbol{U}_{\text{rel}} |\boldsymbol{U}_{\text{rel}}| \tag{14}$$

$$\boldsymbol{\tau}_{\text{D}} = -\,0.5 C_{\text{a}} \rho_w l^3_{avr} A_{sub} \boldsymbol{\Omega} |\boldsymbol{\Omega}| \tag{15}$$

where C_D and C_{a} denote the drag and drag moment coefficients, with the values of 1.0 and 0.05 respectively; $\boldsymbol{U}_{\text{rel}}$ is the relative velocity between the ice floe and the water at the center of buoyancy; $\boldsymbol{\Omega}$ is the angular velocity of the ice floe; l_{avr} is the average characteristic length of the ice floe; A_{sub} is the area of the submerged part of the ice floe; and A_{proj} is the projection area of A_{sub} in the direction of $\boldsymbol{U}_{\text{rel}}$.

3.5. Resistance identification method

In an impulse-based non-smooth time stepping scheme, the rigid-body momentum $\boldsymbol{J}_p$ update over one time step Δt is written in impulse form:

$$m_p \left(\boldsymbol{v}_p^{n+1} - \boldsymbol{v}_p^n \right) = \boldsymbol{J}_p = \lambda_p^{\text{ext}} + \lambda_p^{\text{con}} + \lambda_p^{\text{c}} = \boldsymbol{F}_p \triangle t \tag{16}$$

where λ_p^{ext} is the impulse contribution of external forces (gravity, buoyancy, hydrodynamic drag, towing actuation, etc.), λ_p^{con} is the impulse contribution of contact interaction with ice (normal impact and friction impulses) accumulated over all contacts in this time step, λ_p^{c} is the impulse from kinematic/bilateral constraints used to enforce selected DOFs.

Taking the y-component of Eq. (16) yields:

$$m_p \left(\boldsymbol{v}_{p,y}^{n+1} - \boldsymbol{v}_{p,y}^n \right) = \boldsymbol{J}_{p,y} = \lambda_{p,y}^{\text{ext}} + \lambda_{p,y}^{\text{con}} + \lambda_{p,y}^{\text{c}} = \boldsymbol{F}_{p,y} \triangle t \tag{17}$$

Dividing Eq. (17) by Δt gives the step-averaged total force:

$$\boldsymbol{F}_{p,y} = \boldsymbol{F}_{p,y}^{\text{ext}} + \boldsymbol{F}_{p,y}^{\text{con}} + \boldsymbol{F}_{p,y}^{\text{c}} = \frac{\boldsymbol{J}_{p,y}}{\triangle t} \tag{18}$$

When the y-direction constraint is released, i.e. the plate is not artificially supported/locked by a bilateral constraint in y, then the plate's net force is only from external loading and contact interaction. In that case, the hydrodynamic/actuation resistance can be defined as the negative of the non-contact contributions:

$$\boldsymbol{R}_{p,y} = \boldsymbol{F}_{p,y}^{\text{ext}} + \boldsymbol{F}_{p,y}^{\text{con}} \tag{19}$$

From Eq. (19), rearranging to isolate the non-constraint part yields:

$$\boldsymbol{R}_{p,y}=\boldsymbol{F}_{p,y}-\boldsymbol{F}^{\mathrm{c}}_{p,y}=\frac{\boldsymbol{J}_{p,y}-\lambda^{\mathrm{c}}_{p,y}}{\triangle t}=\frac{m_p\left(\boldsymbol{v}^{n+1}_{p,y}-\boldsymbol{v}^{n}_{p,y}\right)}{\triangle t}-\frac{\lambda^{\mathrm{c}}_{p,y}}{\triangle t} \tag{20}$$

$$\boldsymbol{F}^{\mathrm{con}}_{p,y}=\boldsymbol{R}_{p,y}-\boldsymbol{F}^{\mathrm{ext}}_{p,y}=\frac{m_p\left(\boldsymbol{v}^{n+1}_{p,y}-\boldsymbol{v}^{n}_{p,y}\right)}{\triangle t}-\frac{\lambda^{\mathrm{c}}_{p,y}}{\triangle t}-\boldsymbol{F}^{\mathrm{ext}}_{p,y} \tag{21}$$

In the present simulation, the plate resistance in the y-direction is extracted directly from the stepwise momentum balance, i.e., it is proportional to the inertial term evaluated from the discrete velocity increment, $\boldsymbol{R}_{p,y}$ with the constraint impulse term $\frac{\lambda^{\mathrm{c}}_{p,y}}{\triangle t}$ subtracted when a bilateral/kinematic constraint contributes to the y-direction reaction.

In many DEM/NDEM studies, collision-induced resistance is reported by converting contact impulses from the linear complementarity contact solver into "instantaneous" per-contact forces and then summing/projection over all contacts. For dense stochastic broken-ice interactions, such contact-level force reporting can be sensitive to rapidly evolving contact topology (birth/death of contacts, stick–slip switching, and tolerance-level micro separation/re-contact). The difficulty becomes more pronounced when the target is not only a global resultant but cell-resolved loads needed for sensel-scale pressure reconstruction, because the net load on each cell would require explicit identification and accumulation of the contact/constraint impulses associated with that cell at every time step.

The present approach avoids relying on contact-by-contact force reconstruction for reporting the collision-induced resistance. Instead, the net ice-induced load is identified directly from the time-discrete impulse–momentum balance using the measured state increment of the target body over the time step. Although equivalent in principle to summing contact impulses over the same step, this formulation provides a streamlined, solver-consistent step-averaged load that is more compatible with bandwidth-limited experimental measurements. Crucially, the same momentum-based identification extends naturally to the cell level in Section 3.6, enabling cell-wise net ice loads to be obtained without additional impulse bookkeeping from the linear complementarity contact solver, which directly supports sensel-scale collision-pressure monitoring.

3.6. Pressure identification method

Local pressure is experimentally measured on instrumented structures, but it is difficult to obtain in impulse-based non-smooth DEM because the contact is multi-point, highly intermittent, and rapidly migrating. To address the plate–broken-ice pressure problem, the present study introduces a sensor-consistent pressure identification strategy that explicitly simulates and monitors localized ice-induced collision pressure on the structure within an impulse-based NDEM framework.

To reconstruct a sensor-consistent local pressure field from an impulse-based NDEM solver, the plate face is discretized into M surface cells Ω_m ($m = 1, \ldots, M$) with areas A_m. Each cell is represented by a thin rigid patch whose state variables are x_m, R_m, v_m, ω_m.

To keep the discretized plate effectively non-deforming without introducing stiff inter-cell constraints, the mass and inertia of each cell body are artificially amplified,

$$m_m=\alpha\widetilde{m}_m, I_m=\alpha\widetilde{I}_m, \alpha\gg 1 \tag{22}$$

where $\widetilde{m}_m$, $\widetilde{I}_m$ are the nominal values. In an impulse-based update, the cell velocity jump scales with the net impulse magnitude:

$$\Delta\boldsymbol{v}_m=\boldsymbol{v}^{n+1}_m-\boldsymbol{v}^{n}_m=\frac{\boldsymbol{J}_m}{m_m}, \Delta\boldsymbol{\omega}_m=\boldsymbol{\omega}^{n+1}_m-\boldsymbol{\omega}^{n}_m=\frac{\boldsymbol{L}_m}{I^{-1}_m} \tag{23}$$

Hence, for bounded collision impulses $\boldsymbol{J}_m$, $\boldsymbol{L}_m$ the inertia amplification yields:

$$\|\triangle\boldsymbol{v}_m\|=O\left(\alpha^{-1}\right), \|\triangle\boldsymbol{\omega}_m\|=O\left(\alpha^{-1}\right) \tag{24}$$

so the incremental displacement/rotation per step is $O(\alpha^{-1}\Delta t)$. A sufficiently large α suppresses inter-cell relative motion, allowing the discretized plate to preserve its shape within a controllable tolerance. Here, α is chosen such that the relative displacement between neighboring cells is smaller than 1% of the sensel size throughout the simulation, so the plate behaves as a rigid assembly while still allowing cell-resolved load accounting.

Over one time step Δt, the impulse-momentum update of cell m is written in the same form as the plate-level relation in Eq. (16):

$$m_m\left(\boldsymbol{v}^{n+1}_m-\boldsymbol{v}^{n}_m\right)=\lambda^{\mathrm{ext}}_m+\lambda^{\mathrm{con}}_m+\lambda^{c}_m \tag{25}$$

where λ^{ext}_m is the non-contact external impulse on cell m (gravity, buoyancy, drag, actuation), λ^{con}_m is the ice-cell contact impulse accumulated during the step, and λ^{c}_m is the impulse from possible bilateral/kinematic constraints on cell m.

The step-averaged contact force on cell m is:

$$\boldsymbol{F}^{\mathrm{con}}_m=\frac{\lambda^{\mathrm{con}}_m}{\Delta t}=\frac{m_m\left(\boldsymbol{v}^{n+1}_m-\boldsymbol{v}^{n}_m\right)}{\Delta t}-\boldsymbol{F}^{\mathrm{ext}}_m-\boldsymbol{F}^{c}_m \tag{26}$$

In the present implementation, explicit inter-cell constraints are not used (shape is maintained by inertia amplification), hence $\boldsymbol{F}^{c}_m$ is negligible; moreover, the pressure reconstruction is governed by the contact contribution projected onto the plate normal.

Let $\boldsymbol{n}_p$ denote the unit normal of the plate face. The cell-wise (step-averaged) normal pressure is defined as:

$$p_m=\frac{\boldsymbol{F}^{\mathrm{con}}_m\cdot\boldsymbol{n}_p}{A_m} \tag{27}$$

Accordingly, the area-integrated cell pressures recover the resultant plate-normal collision load.

$$\sum_{m=1}^{M}p_mA_m=\sum_{m=1}^{M}\boldsymbol{F}^{\mathrm{con}}_m\cdot\boldsymbol{n}_p=\boldsymbol{F}^{\mathrm{con}}_p\cdot\boldsymbol{n}_p \tag{28}$$

which provides an explicit conservation check that the reconstructed pressure field is physically meaningful.

With the rigid high-inertia cell-assembly representation introduced above, the plate is effectively converted into a pressure-extraction unit whose 20 × 8 footprint matches the tactile array, so that localized ice–structure collision pressures can be resolved and reported at the sensel scale rather than only as a plate-averaged resultant. To make this local pressure simulation practical in dense stochastic broken-ice contact, the cell-wise net ice load is identified directly from momentum conservation using the cell state increment between two consecutive steps, avoiding the conventional need to track and accumulate large numbers of time-varying contact/constraint impulses for each target cell from the linear complementarity contact solver; although equivalent in principle to impulse summation over the step, the state-increment form provides a simpler, solver-consistent workflow and a direct conversion from cell load to collision pressure through projection and area normalization on the fixed sensor footprint. As a result, to the best of the authors' knowledge, this formulation provides the first practical impulse-based non-smooth DEM implementation that explicitly simulates and monitors localized collision pressure during structure–broken-ice interaction on a sensor-consistent grid. The approach to monitoring localized ice-induced pressure in NDEM simulations has not been published elsewhere.

3.7. Model validation

To build confidence in the underlying impulse-based NDEM setup before applying it to the instrumented plate cases and the associated sensel-scale pressure statistics, an external validation against publicly available ship–broken-ice experiments is first performed. Specifically, the published broken-ice resistance tests of the Xuelong-2 icebreaker

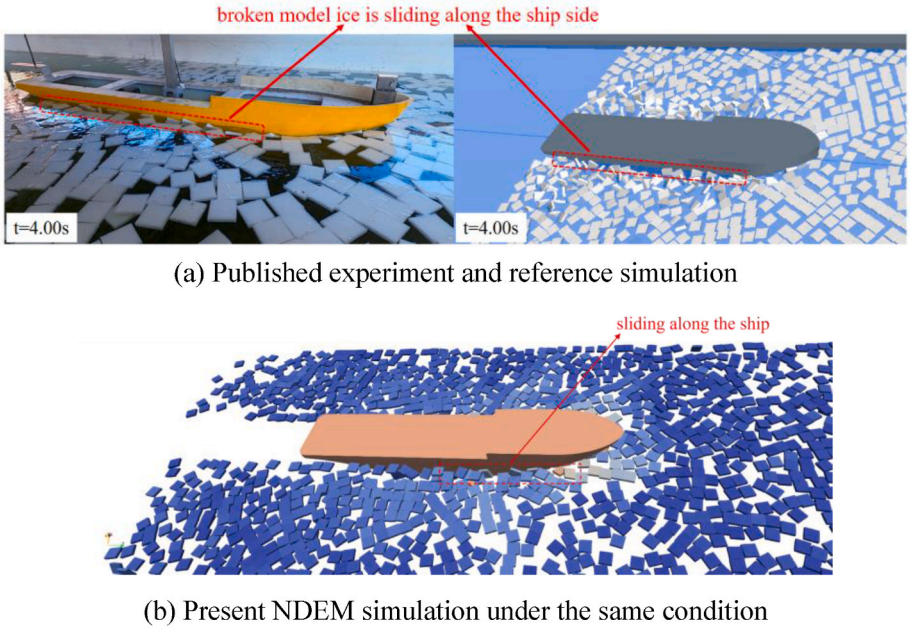


Fig. 10. Comparison of ship-ice interaction snapshots with published data (Xue et al., 2024).

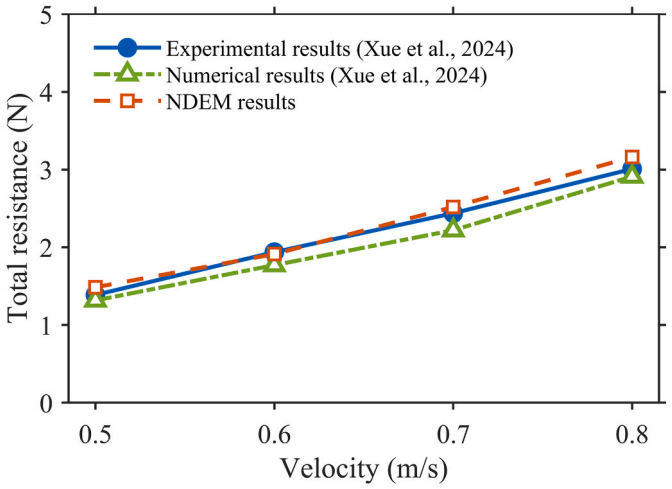


Fig. 11. Total resistance comparison with published experiments (Xue et al., 2024).

Table 1
Total Resistance comparison with published experimental data (Xue et al., 2024).

V_{ship} (m/s)	η	$R_{ref.\ exp}$ (N)	$R_{ref.\ num}$ (N)	R_{NDEM} (N)	Error
0.5	60%	1.386	1.316	1.485	7.14%
0.6	60%	1.937	1.77	1.912	−1.29%
0.7	60%	2.438	2.219	2.519	3.32%
0.8	60%	3.008	2.91	3.161	5.09%

reported by Xue et al. (2024) are reproduced. The experiments were conducted in the towing tank of Harbin Engineering University using polypropylene (PP) model ice. In that campaign, the ship advanced through a broken-ice field with an ice concentration of $\eta = 60\%$ at four towing speeds, V_{ship} = 0.5, 0.6, 0.7, and 0.8 m/s, and corresponding reference numerical predictions were also provided for the same conditions. Following the published setup, an equivalent NDEM model (geometry, broken-ice field configuration, and towing conditions) was constructed and simulations were performed for the same speed series.

Fig. 10 compares representative ship–broken-ice interaction snapshots. The present NDEM reproduces the main near-field interaction feature reported in the published study, namely the model-ice pieces sliding along the ship side, indicating that the dominant kinematic mode and contact evolution are captured consistently with the reported experiment and reference simulation.

Quantitative validation is presented in Fig. 11 and Table 1, where the total resistance predicted by the present NDEM is compared with the published experimental measurements and the reported reference numerical results. The present NDEM captures the same monotonic increase of resistance with towing speed and yields resistance levels close to the published experiments for all four cases. The relative deviation from the published experimental resistance ranges from −1.29% to 7.14%, and the mean absolute percentage error is 4.21%. Overall, these comparisons support that the present impulse-based NDEM setup and the proposed momentum-conservation-based load identification procedures provide reliable resistance predictions under published broken-ice test conditions.

3.8. Computational domain set up

After the above external validation, the NDEM setup for the present instrumented-plate simulations is described in this section. The computational domain replicates the towing-tank conditions in terms of geometry, broken-ice extent, and prescribed plate motion. Following the experimental post-processing window, the computational domain represents the 10 m × 2 m broken-ice test section only. For η = 40%, 60% and 80%, square floes were randomly placed and manually redistributed to obtain an approximately uniform initial areal coverage (Fig. 12). Rigid lateral boundaries were applied to mimic the channel confinement, while the downstream end was kept open to mitigate non-physical accumulation due to boundary reflection. The coordinate system is consistent with the experiments (Y towing, X transverse, Z upward).

The broken ice field was modeled as an assembly of rigid square floes with a nominal size of 0.067 × 0.067 × 0.015 m, matching the model ice pieces used in the tests. The floe density was set to ρ_i = 917.0 kg/m^3, within the measured range of the polypropylene (PP) material (890.0-917.0 kg/m^3), while the water density was set to ρ_w = 1000.0 kg/m^3. Coulomb friction was adopted for all contacts, and the kinetic friction

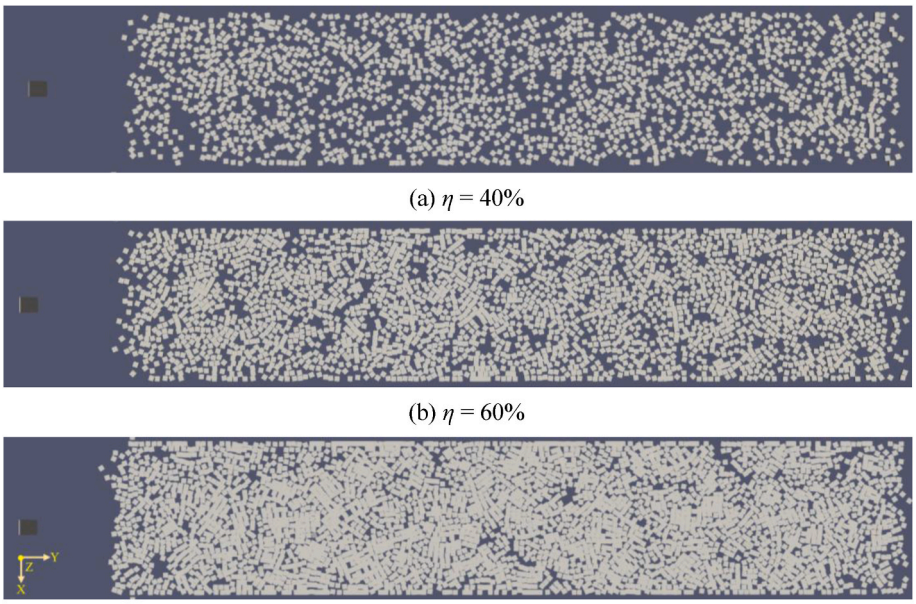


Fig. 12. The computation domain before the plate enters the ice floe region.

Table 2
Parameters of the NDEM simulation (model scale).

Parameters	Values
Water density(kg/m^3)	1000.0
Ice density(kg/m^3)	917.0
Ice thickness(m)	0.015
Ice-plate friction coefficient	0.138
Ice-ice friction coefficient	0.29
Ice- plate restitution coefficient	0.5
Ice-ice restitution coefficient	0.25
Time step (s)	0.0005
Form drag coefficient C_d	1.0
Angular drag coefficient C_a	0.05

coefficients were prescribed as $\mu_{i\text{-}p} = 0.138$ for the wet plate–ice interface and $\mu_{i\text{-}i} = 0.29$ for ice-ice contacts. Importantly, these two coefficients were obtained from dedicated friction measurements performed in the present experimental setup, and the identical values were then used in the numerical simulation for consistency. These values are also within commonly reported ranges of ship-ice and ice-ice kinetic friction in the literature (Timco and Weeks, 2010; Zhong et al., 2023), thereby ensuring a like-for-like replication of the sliding response observed in the model tests. The material/contact parameters, fluid-force coefficients, and the time step are summarized in Table 2.

The plate was modeled as rigid plate assembly (0.20 × 0.25 × 0.02 m) under a fixed-waterline condition (0.15 m submerged and 0.10 m emerged), identical to the experiments. The inclination angle $\theta =$ 30°, 45°, 60°, 90° and towing speed $V = 0.50$, 0.59, and 0.68 m/s follow the test matrix. To mimic the experimental constraints, the plate was restricted to translation along the towing direction, with lateral motion and rotation suppressed.

For direct comparison with the tactile measurements, the plate face was discretized into a 20 × 8 array (160 cells) matching the instrumented sensel layout. Each cell corresponds to an in-plane size of approximately 10 × 33 mm and is represented by a rigid patch element, enabling impulse-based pressure reconstruction at the sensel scale while maintaining a rigid overall plate.

4. Results and discussion

4.1. Ice-plate interaction

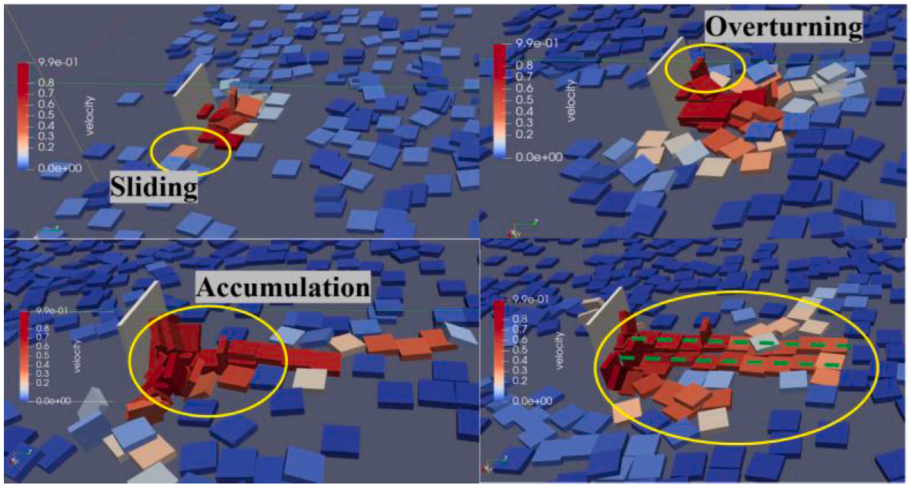
Under the reference condition $\eta = 60\%$, $\theta = 90°$, $V = 0.5$ m/s, the experiments and NDEM simulations show the same dominant near-field interaction modes (Fig. 13): (i) side sliding/bypassing, (ii) local overturning near the frontal stagnation zone, (iii) intermittent frontal accumulation (local jamming), and (iv) wake-induced streamwise “ice chains”. While the exact timing varies between runs due to random floe arrangements, these modes are consistently observed in both datasets. Most contacts occur as short-lived edge/corner engagements that redirect floes laterally along the plate sides, forming a bypassing stream that clears the immediate upstream region. Overturning events are repeatedly triggered by eccentric frontal impacts that generate a moment on a floe, after which the floe either briefly travels with the plate or rapidly joins the lateral bypassing flow. In contrast, frontal accumulation is intermittent but crucial: under severe accumulation, successive impacts can temporarily interlock a floe cluster ahead of the plate, so the plate pushes a connected group rather than isolated pieces, potentially forming transient ice force chains and producing a short-lived strong-interaction state.

A representative formation-collapse sequence of this collective state is shown in Fig. 14: a loosely distributed upstream field transitions into a compacted, load-bearing contact network (“force-chain-like” transmission) and then rapidly collapses as contacts rearrange and floes rotate/escape. This transient collective loading provides the physical basis for the burst-like resistance events reported below.

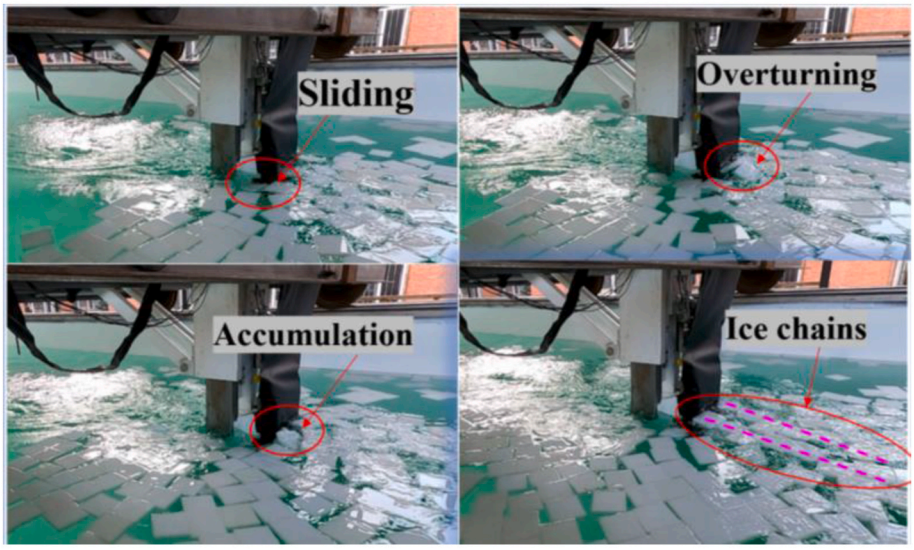
At the basin scale, the simulations reproduce the lateral clearing around the trajectory and the wake redistribution pattern observed in the experiments (Fig. 15), supporting that the model captures both the local interaction modes and the resulting ice-field reorganization.

4.2. Results of resistance

Under $\eta = 60\%$, $\theta = 90°$, $V = 0.5$ m/s, the plate resistance was obtained by summing the four load-cell signals and taking the component along the towing direction. Fig. 16 shows a representative record over the full run and is divided (by dashed lines) into four stages. In Stage 1 (acceleration), the measured resistance rises rapidly because the



(a) NDEM simulation snapshots



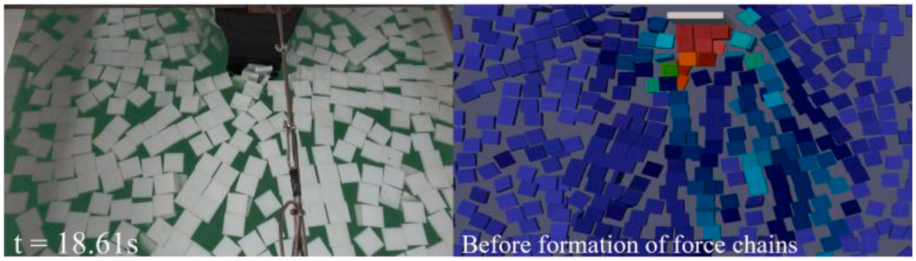
(b) Corresponding experimental observations

Fig. 13. Typical interaction modes between a vertical plate and broken ice floes ($\eta = 60\%$, $\theta = 90°$, $V = 0.5$ m/s).

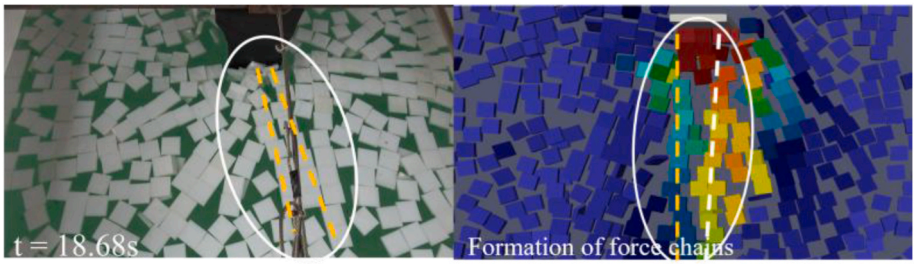
carriage and plate cannot reach the target speed instantaneously due to inertia. Stage 2 corresponds to steady towing in open water, where the resistance remains at a low quasi-steady level with only small fluctuations, representing the plate's hydrodynamic towing resistance. In Stage 3 (broken-ice interaction), the resistance becomes strongly intermittent: a quasi-stationary background is punctuated by frequent sharp bursts and occasional extreme peaks, reflecting repeated transitions between sliding/bypassing-dominated contacts and transient frontal accumulation (local jamming) of floes. Finally, in Stage 4 (deceleration after leaving the ice field), the signal drops quickly and may briefly exhibit negative components during braking before returning toward the baseline.

It should be noted that the NDEM plate is driven with a prescribed constant towing velocity; therefore, the numerical resistance history does not include time-resolved acceleration/deceleration stages as in the experiment. Once motion starts, the simulated resistance immediately reaches an open-water level dominated by the hydrodynamic drag term computed using Morison's equation (Eq. (14)) and remains nearly constant until ice contacts occur. Fig. 17 compares the resistance time histories over the broken-ice segment between experiment and simulation for $\eta = 60\%$, $\theta = 90°$ and $V = 0.5$ m/s. The experiment gives a mean total resistance of 7.544 N and a mean ice-induced component of 1.734 N after subtracting the open-water baseline. The NDEM prediction is 6.742 N (total) and 1.892 N (ice-induced), corresponding to deviations of -10.6% and $+9.1\%$, respectively. Although individual peaks do not align in time due to configuration-dependent stochasticity in a broken-ice field, the simulation reproduces the characteristic intermittency and a comparable spike envelope during Stage 3, indicating that the dominant peak-generating processes in broken-ice interaction are captured.

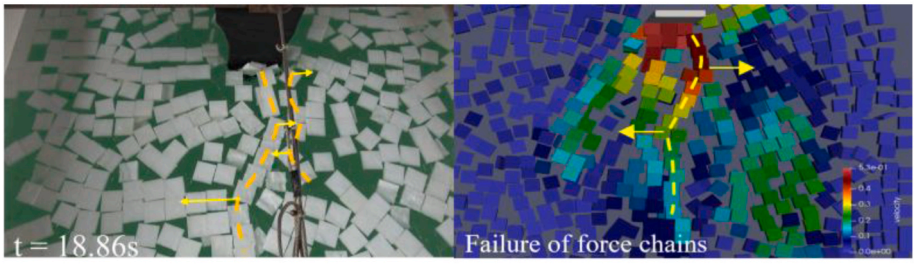
The peak-dominated statistics are quantified in Fig. 18 using exceedance probability curves $P(F > x)$ for $\eta = 40\%$, 60% and 80% at $\theta = 90°$. The exceedance probability decreases rapidly with resistance, indicating that low-to-moderate levels dominate and large instantaneous events are rare. Increasing η shifts the exceedance curves to higher resistance and yields a systematically longer tail. This concentration ranking is clear in the tail region: $\eta = 80\%$ exhibits the slowest decay (most persistent heavy tail), $\eta = 60\%$ is intermediate, and $\eta = 40\%$ drops to negligible exceedance at much smaller x. The NDEM results reproduce the same ordering and tail trends, supporting the model's ability to predict not only mean resistance levels but also the probability of extreme instantaneous events.



(a) t = 18.60 s



(b) t = 18.68 s



(c) t = 18.86 s

Fig. 14. Force chain phenomena observed in experiments and simulations (η = 60%, θ = 90°, V = 0.5 m/s).

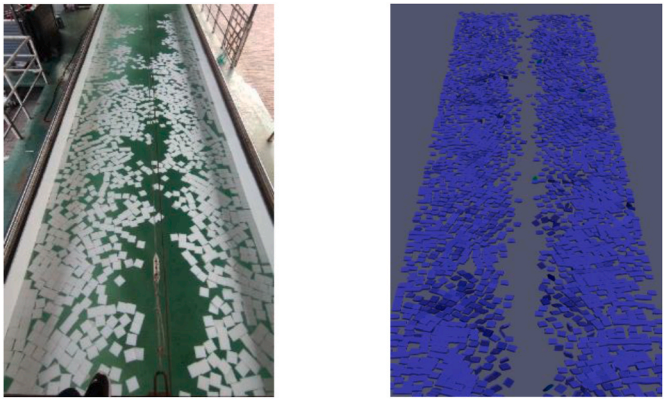


Fig. 15. Ice-channel pattern after the passage of the plate (η = 60%, θ = 90°, V = 0.5 m/s): comparison between experiment (left) and NDEM simulation (right).

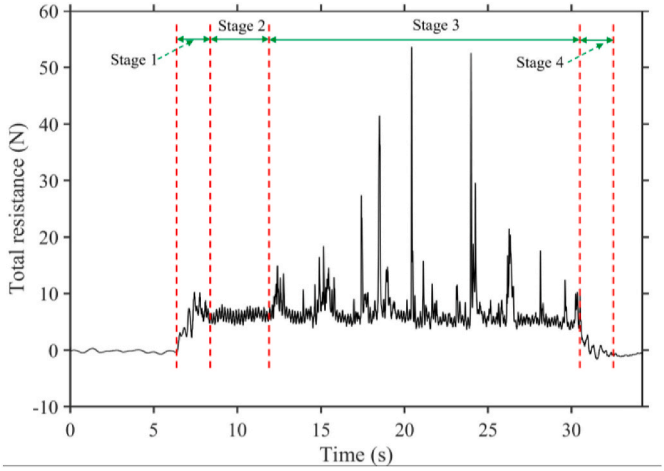


Fig. 16. Time history curves of total resistance at η = 60%, θ = 90°, V = 0.5 m/s.

4.3. Results of pressure

Figs. 19–22 show that the measured plate pressure is strongly localized in space and intermittent in time, and that the NDEM results reproduce these features at the level relevant to hot-spot location and global statistics. At the instant of the global peak, the pressure field concentrates in a single short-lived patch rather than a distributed load (Fig. 19(a, c)). Over the entire traversal, the per-sensel maximum map further reveals a persistent hot-spot band near the waterline and its adjacent submerged region, whereas the emerged region shows

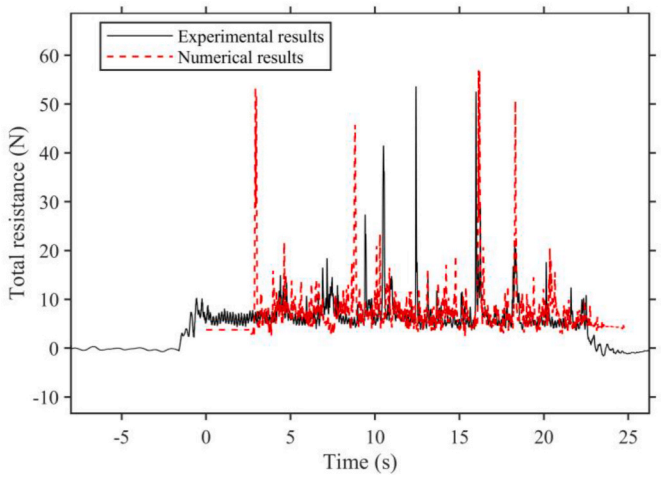


Fig. 17. Time history of the plate total resistance: comparison between experiment and NDEM simulation.

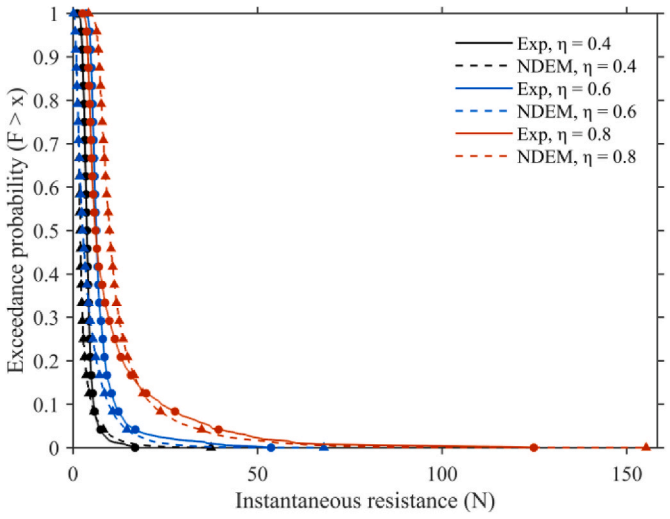


Fig. 18. Exceedance probability of instantaneous ice resistance at different ice concentrations.

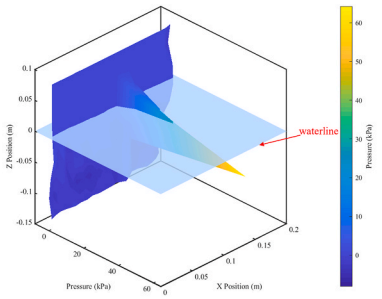
markedly lower maxima (Fig. 19(b, d)). Compared with the experiment, the NDEM map exhibits a slightly broader footprint of elevated maxima, consistent with a more spatially distributed sequence of moderate contacts while retaining the same waterline-dominated hot-spot pattern.

The intermittency is clearly reflected in the maximum-pressure histories (Fig. 20): both records are dominated by short-lived bursts that are separated by comparatively quiet intervals. To enable a like-for-like temporal comparison, the NDEM pressure series was downsampled to 50 Hz to match the Tactilus sampling rate, which reduces artificial frame-to-frame fluctuations caused by the much smaller simulation time step. However, the NDEM series still contains a noticeably higher density of moderate peaks throughout the active interaction window, whereas the experimental trace shows fewer, more isolated burst events with a flatter background level. With the sampling rate now matched, this remaining difference is better attributed to physics and the measurement chain rather than temporal resolution. In the impulse-based NDEM, rapidly evolving micro-contacts, including frequent brief touch-separate-recontact events and intermittent stick-slip transitions, are resolved and directly reported through the impulse-to-cell pressure reconstruction, so many weak impacts remain visible as distinct local maxima. In contrast, the tactile sensing chain introduces attenuation and smoothing through the compliant sensor layers, finite dynamic response, spatial averaging and cross-talk between neighboring sensels, and the baseline correction procedure, which can also produce small negative offsets in the quiescent stage. It is also noted that the time of the maximum local pressure does not need to coincide with the time of the maximum global towing resistance. The towing resistance is a global resultant in the towing direction aggregated over all concurrent contacts and frictional interactions, and it typically peaks during short-lived multi-floe accumulation or jamming episodes when a temporarily load-bearing contact network forms ahead of the plate. In contrast, the maximum pressure is a highly localized normal extreme that can be triggered by an intense edge or corner crushing event over a small patch of the plate, even when the global contact network is not at its strongest. Therefore, a time offset between the maxima of the two signals is physically expected in stochastic broken-ice interaction.

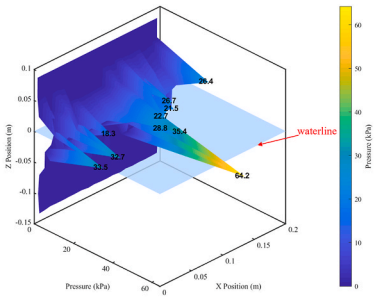
To visualize where extreme events concentrate on the plate, Fig. 21 presents spatiotemporal maxima surfaces obtained by taking directional envelopes of the 20 × 8 tactile array. In Fig. 21(a), the plate is discretized into 20 widthwise columns; at each time step, the pressure in a given column is defined as the maximum of the eight sensels along the vertical direction, yielding the column-wise maximum field $p_{\max}(x,t)$, which highlights the temporal evolution and lateral localization of high-pressure patches along the plate width. By contrast, in Fig. 21(b), an orthogonal reduction is performed: the plate is discretized into 8 vertical rows, and the pressure in each row is taken as the maximum of the 20 sensels across the width, resulting in the row-wise maximum field $p_{\max}(z,t)$, which emphasizes where extremes occur along the vertical coordinate relative to the waterline. The column-wise maxima $p_{\max}(x,t)$ are characterized by sparse, spike-like bursts confined to limited width locations, indicating that extreme pressures are highly localized rather than laterally distributed. By contrast, the row-wise maxima $p_{\max}(z,t)$ show that the largest and most frequent extremes occur around the waterline and the adjacent submerged rows, consistent with intermittent impacts and short-lived compaction occurring in this near-surface band.

Fig. 22 complements this global vertical view by focusing on the waterline vicinity and separating it into three bands (waterline, above-water, and submerged). The waterline band exhibits the highest event density and sustained intermittency, reflecting repeated confinement, redirection, and intermittent local jamming of floes in the frontal region. The above-water band generally remains at low levels and features fewer pronounced events, suggesting that direct ice loading there is less frequent under the fixed-waterline configuration, although sporadic peaks can still occur when floes climb, overturn, or ride up along the plate. The submerged band is characterized by fewer events but more pronounced isolated spikes, highlighting that rare below-water contacts can generate high local pressures even though the dominant loading process is concentrated near the waterline.

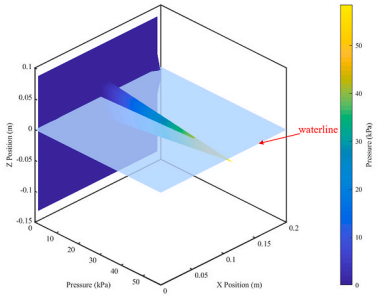
To separate the low-probability, representative broken-ice impacts from the quasi-continuous water loading and minor ice contacts, the pressure records obtained in the broken-ice runs were first referenced to the open-water stage. Specifically, for each sensel, the mean pressure measured during steady towing in pure water was subtracted, so that the incremental signal $\Delta p(t)$ represents the pressure deviation relative to the open-water baseline, effectively removing sensor offset and the quasi-static hydrostatic contribution at that location. The probability distribution of Δp was then evaluated by pooling samples from all sensels, yielding a global mean μ_{all} and standard deviation σ_{all} for the mixed population dominated by water-induced fluctuations with intermittent collision-induced excursions. Since water-related loading persists for almost the entire record, while strong ice impacts occur only intermittently, the pooled Δp samples are largely concentrated in the low-pressure range. Only a very small fraction exhibits pronounced high-Δp excursions that deviate markedly from the bulk distribution, and these rare events are attributed to distinct broken-ice collisions with the plate. Therefore, a separation threshold $u = \mu_{all} + 3\sigma_{all}$ was adopted to extract



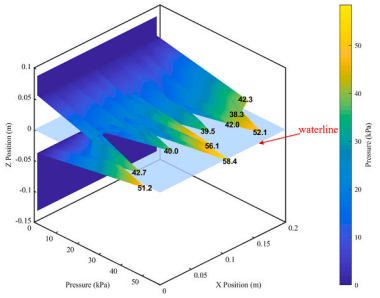
(a) Experimental pressure field at the time of the global peak pressure



(b) Experimental map of per-cell maximum pressure during the test

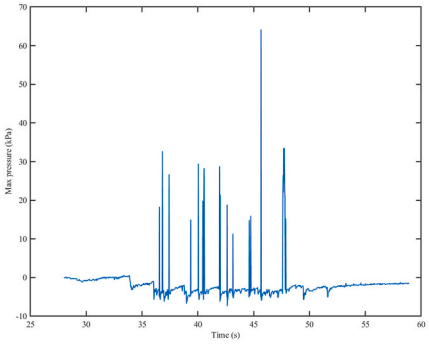


(c) Numerical pressure field at the time of the global peak pressure

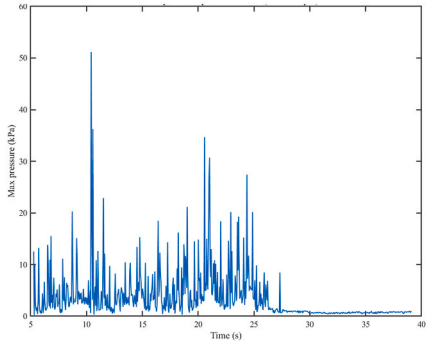


(d) Numerical map of per-cell maximum pressure during the simulation

Fig. 19. Spatial distribution of plate pressure in experiments and simulations.

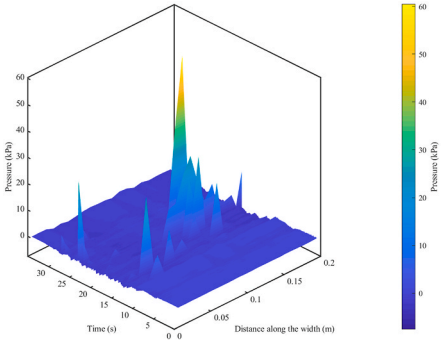


(a) Experiment (maximum among all sensing cells)

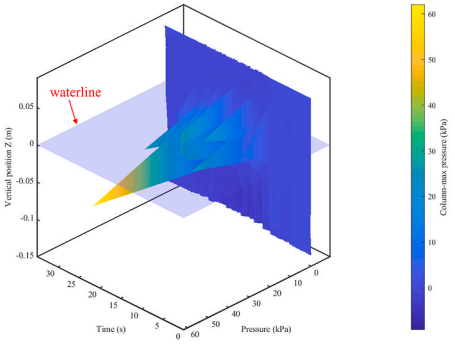


(b) NDEM simulation (maximum among all sensing cells)

Fig. 20. Time histories of the plate maximum pressure.

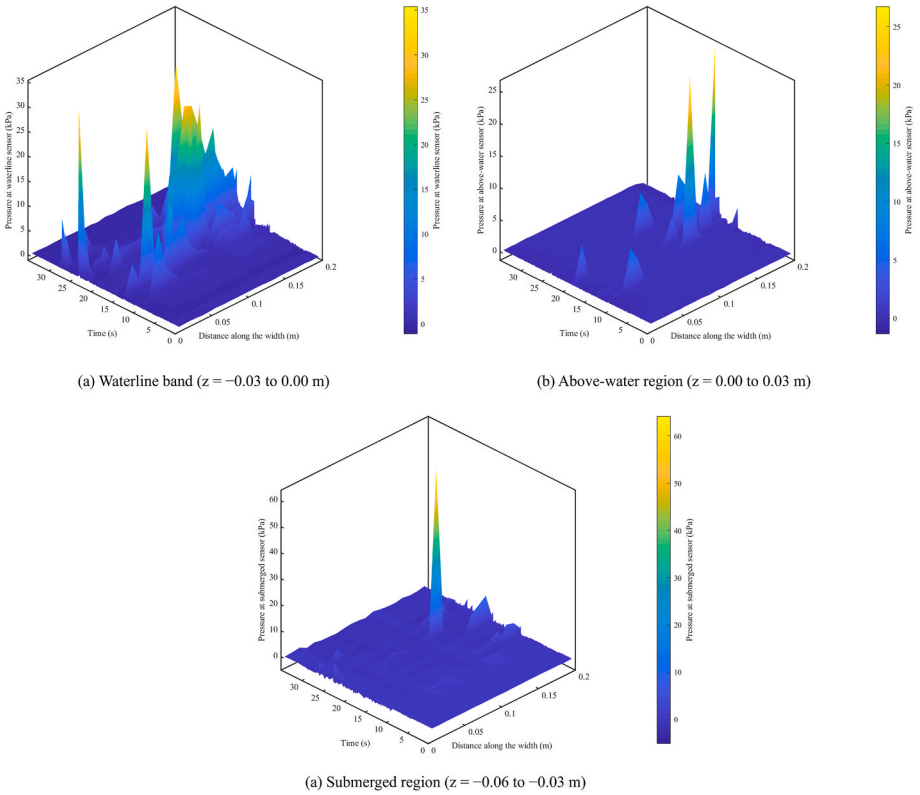


(a) Widthwise spatiotemporal map of column-maximum pressure



(b) Vertical spatiotemporal map of row-maximum pressure

Fig. 21. Spatiotemporal distributions of maximum pressure on the plate measured in the experiment.



(a) Waterline band (z = −0.03 to 0.00 m)

(b) Above-water region (z = 0.00 to 0.03 m)

(a) Submerged region (z = −0.06 to −0.03 m)

Fig. 22. Spatiotemporal pressure distribution near the waterline (experiment).

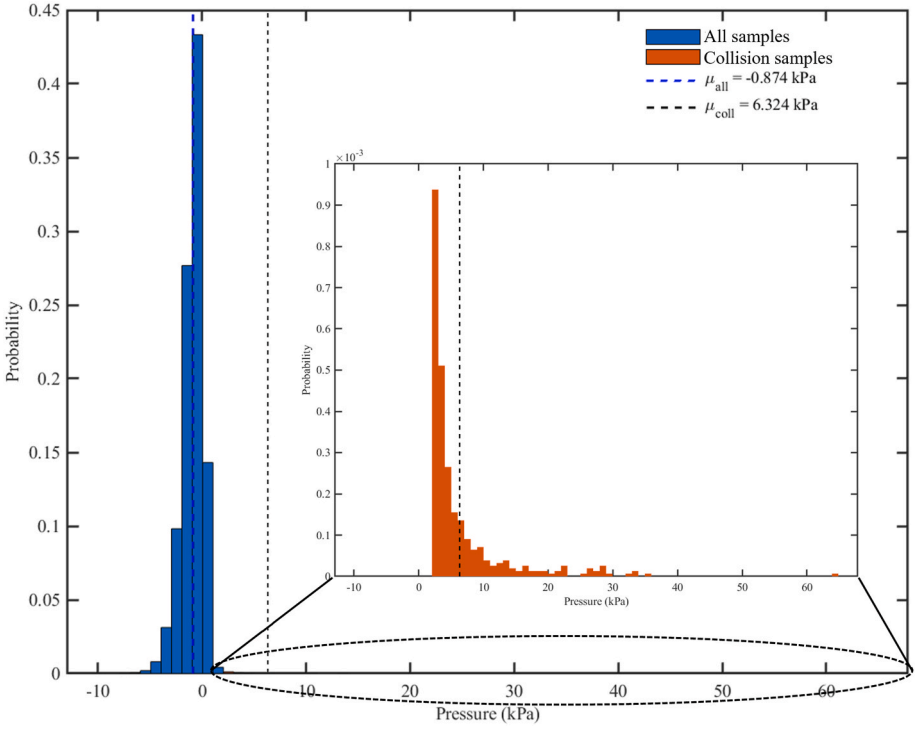
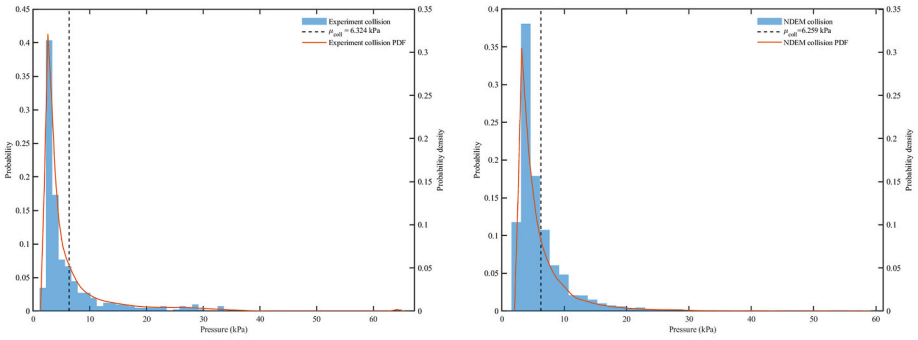


Fig. 23. Probability distribution of plate pressure in the experiment: all samples versus collision samples.

the extreme upper tail. Samples satisfying $\Delta p > u$ were classified as typical broken-ice collision events, and the corresponding samples were retained as the collision-pressure ensemble for subsequent probabilistic analysis. In contrast, samples with $\Delta p < u$ were treated as low-pressure events, which include water-pressure fluctuations together with weak/short-lived ice contacts; in this range, the contributions from water loading and minor ice interactions are difficult to distinguish based on pressure magnitude alone. With this criterion, the retained collision samples account for less than ~0.165% of the full record, indicating that the extracted excursions are predominantly associated with distinct ice-



(a) Experimental collision-pressure distribution (b) Numerical collision-pressure distribution

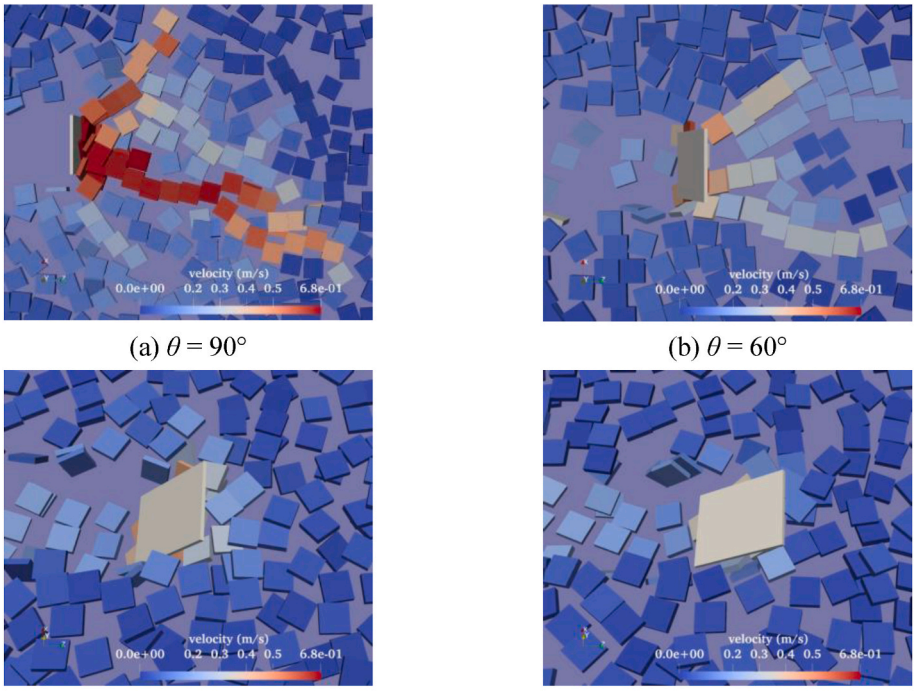
Fig. 24. Statistical characteristics of collision pressure on the plate: comparison between experiment and NDEM simulation.

plate impacts rather than the quasi-continuous water-related loading.

Fig. 23 illustrates the resulting separation between the pooled Δp distribution (all samples) and the extracted collision subset ($\Delta p > u =$ 2.182 kPa). The all-samples histogram is narrowly clustered around small values, with a slightly negative mean ($\mu_{all} = -0.874$ kPa) and a modest spread ($\sigma_{\text{all}} = 1.019$ kPa), indicating that most sensel readings fluctuate weakly about the open-water reference and remain well below the separation threshold. In contrast, the collision subset forms a low-probability right tail with a positive mean ($\mu_{coll} = 6.324$ kPa) and occasional extreme events reaching $O(10^1\text{-}10^2)$ kPa, as highlighted by the inset. The frequent negative Δp values in the low-pressure range ($\Delta p < u$) do not imply tensile loading; rather, they reflect local pressure reductions relative to the open-water baseline caused by flow redistribution around the advancing plate, consistent with the "low-pressure phenomenon" reported in broken-ice and continuous icebreaking conditions (Puntigliano, 2003; Zhong et al., 2023).

Fig. 24 compares the probability characteristics of the extracted typical collision-pressure ensemble between the experiment and the NDEM simulation. To ensure consistency with the separation procedure in Fig. 23, the same threshold obtained from the experimental mixed population, $u = 2.182$ kPa, was also applied to the NDEM pressure series; samples with $\Delta p > u$ were retained as typical collision-induced pressures for the histogram/PDF construction. As shown in Fig. 24(a and b), both datasets exhibit a strongly right-skewed distribution: the probability mass clusters in the low-pressure bins just above the threshold and decays rapidly with increasing pressure, while a sparse heavy tail persists, indicating that most contacts are short-lived grazing/pressing events during sliding/bypassing and that rare intense impacts still occur intermittently, likely linked to transient frontal compaction and force-chain/ice-chain-like load transfer through a temporarily interlocked floe cluster. Quantitatively, the mean collision pressure is $\mu_{\text{coll}} = 6.324$ kPa in the experiment and $\mu_{\text{coll}} = 5.260$ kPa in NDEM (relative difference -16.82%); nevertheless, the overall histogram shape and PDF agree well, suggesting that the impulse-based NDEM model captures the dominant probabilistic structure of collision-induced pressures, including the steep low-to-moderate decay and the persistence of the low-probability high-pressure tail.

Importantly, the event-based separation reveals that the collision-pressure ensemble occupies only a very small fraction of the full pressure record, yet it governs the heavy tail and extreme values that are



(a) $\theta = 90°$ (b) $\theta = 60°$

(c) $\theta = 45°$ (d) $\theta = 30°$

Fig. 25. Influence of inclination angle on ice-plate interaction.

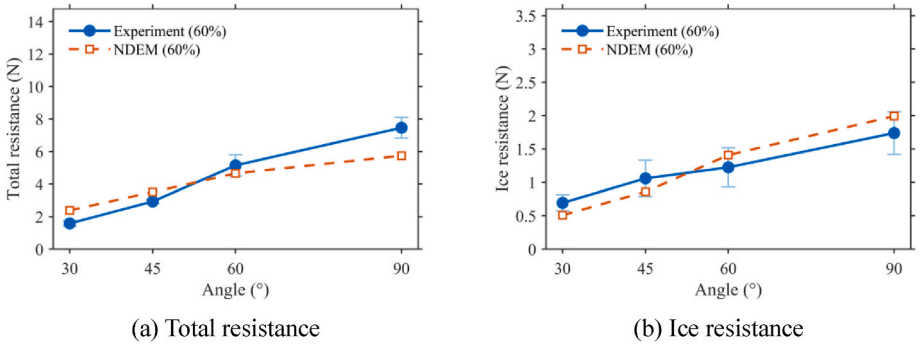


Fig. 26. Total resistance and ice resistance versus angle at $V = 0.5$ m/s and $\eta = 60\%$.

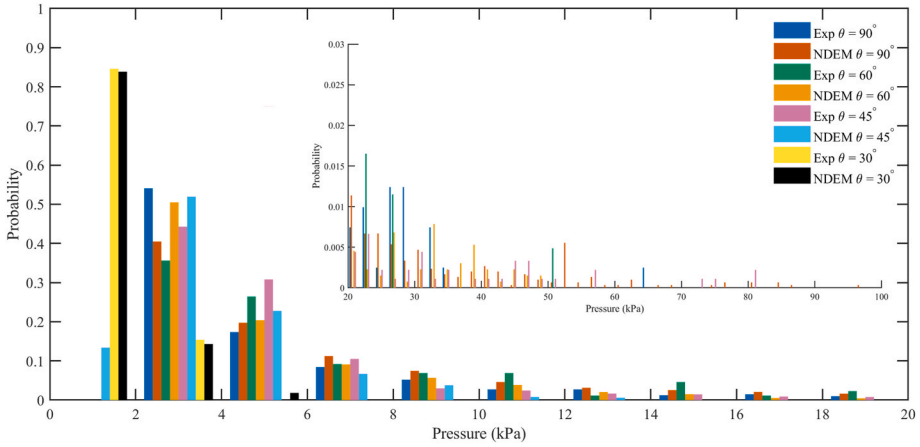


Fig. 27. Collision-induced pressure statistics for different angles at $V = 0.5$ m/s and $\eta = 60\%$.

critical for structural design. This quantifies why mean-pressure or mixed-population statistics can be misleading when water-pressure-dominant samples overwhelm the record. Combined with the persistent waterline-adjacent hot-spot band observed in maxima maps and spatiotemporal envelopes, the present results provide a clear engineering interpretation: extreme loads in broken-ice interaction are controlled by rare, highly localized collision events and their intermittent transition into short-lived accumulation/jamming states, rather than by the quasi-continuous background loading.

Beyond the present like-for-like validation against our towing-tank measurements, the extracted pressure localization and probabilistic signatures are consistent with representative tactile/pressure-sensor studies on ice–structure interactions reported in the literature (Kujala and Arughadhoss, 2012; Sun and Huang, 2020, 2021; Zhong et al., 2023). In particular, prior measurements commonly report strong spatial non-uniformity with localized hot spots and highly intermittent peaks, and statistical analyses indicate that rare collision episodes dominate the upper tail relevant to extreme loads (Suominen and Kujala, 2014; Shamaei et al., 2020; Wu et al., 2021). The present fixed-waterline plate tests and the sensor-consistent NDEM reconstruction reproduce the same key features (waterline-adjacent concentration of extremes and persistent heavy tails). We note that direct one-to-one numerical comparison across different published campaigns is often limited by differences in geometry, sensor footprint/resolution, ice material, and test protocols; therefore, the present work emphasizes strict internal consistency (same geometry and sensing layout) while demonstrating agreement in the dominant physical/statistical signatures reported in the broader literature.

4.4. Sensitivity study

4.4.1. The effect of inclination angle

Figs. 25–26 show that the inclination angle θ primarily regulates the interaction regime and therefore the resistance level. In the simulation snapshots (Fig. 25), the vertical plate ($\theta = 90°$) exhibits a pronounced upstream compaction/stagnation zone and an extended disturbed wake, whereas decreasing θ progressively weakens the upstream compaction and promotes smoother lateral bypassing/deflection. Consistently, at $\eta = 60\%$ and $V = 0.5$ m/s, both total and ice resistance increase monotonically with θ (Fig. 26). The separation between total resistance and ice resistance becomes more evident at large θ, implying that the hydrodynamic towing component (pressure drag associated with a larger effective frontal blockage and the water entrainment by compacted floes) grows increasingly important as the plate approaches the vertical configuration.

Consistent with the monotonic increase in resistance with θ, the collision-pressure probability distributions in Fig. 27 exhibit a systematic shift toward broader ranges and more persistent upper tails as the plate becomes steeper. At $\theta = 30°$, the probability mass is overwhelmingly concentrated in the lowest pressure bins and becomes negligible above 6 kPa in both the experiment and NDEM, indicating that most collision samples correspond to near-threshold events. At $\theta = 45°$, the distribution remains low-pressure-dominated but spreads into the mid-pressure bins, implying a higher occurrence of moderate collision pressures. A clear tail development is observed at $\theta = 60°$ and 90°: the probability decay becomes markedly slower and the high-pressure tail persists to much larger values, with $\theta = 90°$ exhibiting the heaviest tail and thus the highest likelihood of extreme pressures. Overall, the NDEM results reproduce the same angle ranking and tail extension across all cases, showing good consistency with the measurements in

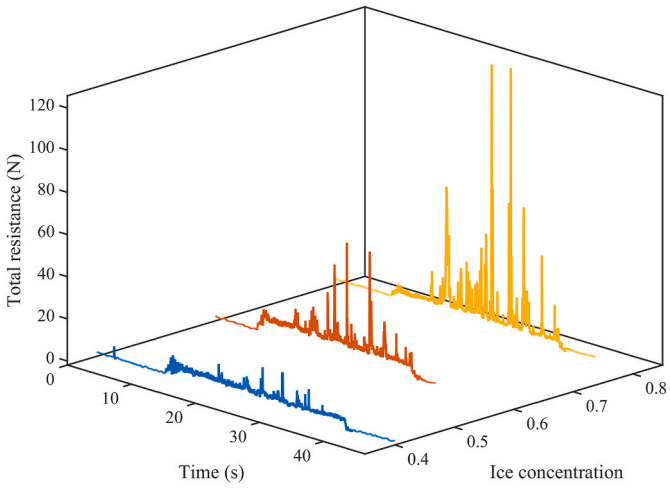


Fig. 28. Time history curves of total resistance at $V = 0.5$ m/s, $\theta = 90°$.

terms of both the dominant low-pressure population and the evolution of rare high-pressure events with increasing θ.

4.4.2. The effect of ice concentration

Fig. 28 indicates a clear concentration dependence of resistance intermittency at $\theta = 90°$: at $\eta = 40\%$, the record remains close to a low quasi-steady level with only occasional spikes; at $\eta = 60\%$, bursts become more frequent and the peak envelope increases; at $\eta = 80\%$, the response becomes strongly peak-dominated with repeated high-amplitude events.

Fig. 29 shows that increasing η progressively constrains lateral bypassing and promotes a denser upstream interaction region, which provides the geometric precondition for more frequent short-lived strong-interaction episodes. The mean trends in Fig. 30 further confirm that both total and ice resistance increase with η, and that the sensitivity is stronger for the ice-induced component; the experimental scatter also increases with η, consistent with the increasingly stochastic occurrence of burst events.

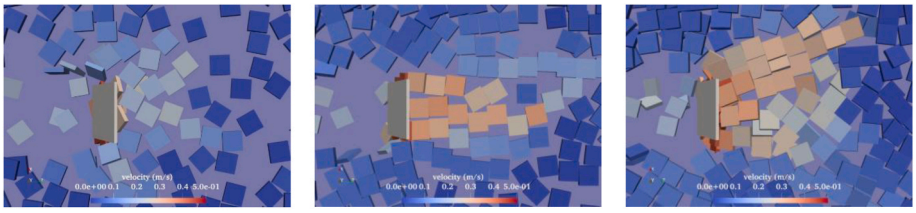
More importantly for local loading, Fig. 31 shows that collision-pressure distributions remain strongly right-skewed for all concentrations: a dominant low-pressure population (typically in the 2-6 kPa range) is accompanied by a sparse extreme tail. With increasing η, probability mass spreads more evidently into the mid-pressure bins and the heavy tail becomes more persistent, indicating that intense collision events become both stronger and more likely in denser floe fields. At $\eta = 40\%$, ample open water space enables rapid release after first contact, so high pressures are strongly suppressed and the tail decays rapidly. At $\eta = 60\%$, reduced bypassing space increases contact multiplicity and re-contact probability, producing more moderate-to-high pressure samples. At $\eta = 80\%$, lateral escape is strongly constrained and transient frontal compaction is more readily formed; while the low-pressure population remains dominant due to numerous weak micro-contacts, intermittent local jamming can concentrate the interaction onto limited patches and extend the extreme tail. The NDEM results capture the same tail-lengthening trend with η, supporting that the concentration dependence of extreme pressures is governed primarily by the probability of entering such transient strong-contact states.

4.4.3. The effect of plate velocity

The collision-pressure distributions in Fig. 32 remain strongly right-skewed across all speeds. The low-to-moderate range shows substantial overlap among $V = 0.50$, 0.59 and 0.68 m/s, implying that the dominant population is still composed of numerous near-threshold impacts and brief pressing contacts. In contrast, the speed effect becomes apparent in the heavy tail: the high-pressure tail becomes more populated and extends to higher pressures as V increases, indicating an increased likelihood of localized intense collision episodes. Physically, a higher approach velocity raises the impulse level per encounter and reduces the time available for floes to rotate/slide away after first contact, which promotes rapid re-contacts and short-lived compacted pressing events near the frontal region, thereby generating rare but stronger pressure peaks.

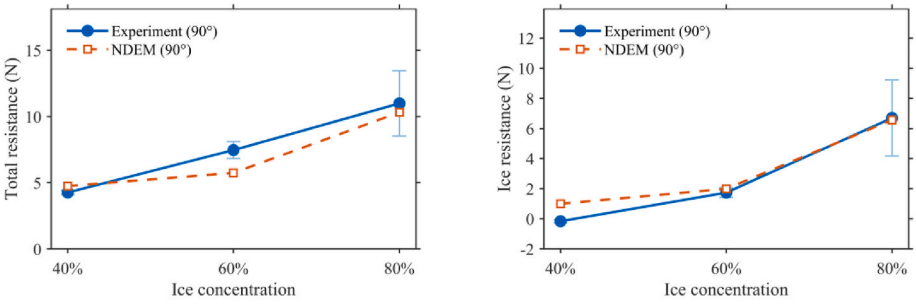
5. Conclusions

For an inclined plate advancing in floating broken ice, towing-tank



(a) η = 40% (b) η = 60% (c) η = 80%

Fig. 29. Influence of concentration on ice-plate interaction at $V = 0.5$ m/s and $\theta = 90°$.



(a) Total resistance (b) Ice resistance

Fig. 30. Total resistance and ice resistance versus ice concentration at $V = 0.5$ m/s and $\theta = 90°$.

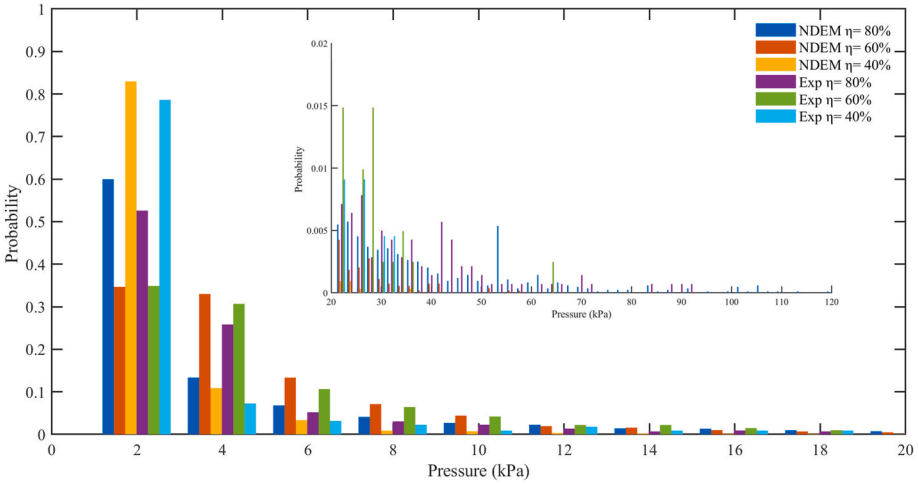


Fig. 31. Collision-induced pressure statistics for different ice concentrations at $V = 0.5$ m/s and $\theta = 90°$.

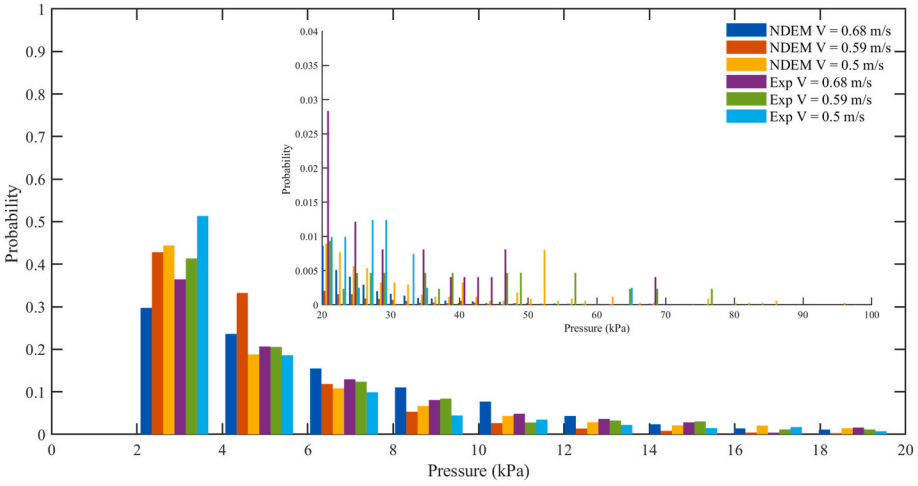


Fig. 32. Collision-induced pressure statistics for different velocities at $\eta = 60\%$ and $\theta = 90°$.

experiments and an impulse-based NDEM simulation were conducted to jointly quantify global resistance and localized collision pressure. A sensor-consistent pressure-extraction methodology was introduced and validated. The main conclusions are as follows:

1. Under a fixed-waterline configuration, towing-tank tests jointly measured global towing resistance and sensel-scale normal pressures on an inclined plate using a 20 × 8 Tactilus sub-array. The near-field interaction was consistently dominated by lateral sliding/bypassing with recurrent local overturning and intermittent frontal accumulation (local jamming), while pressure peaks remained highly intermittent and strongly localized, concentrating around the waterline and the adjacent submerged band.
2. To characterize localized ice-induced pressure, an explicit approach was developed for monitoring collision pressure in impulse-based NDEM simulations of a rigid plate advancing in broken ice. The plate was discretized into a rigid inertia-amplified 20 × 8 cell assembly so that relative cell positions remain essentially unchanged while enabling cell-resolved load accounting; cell loads were obtained directly from the impulse–momentum balance of discrete state increments, avoiding per-contact impulse identification and summation in rapidly evolving stochastic contacts. Representative cases show good agreement in both resistance level and collision-pressure statistics.
3. The measurements and reconstructed pressure fields indicate that ice-induced pressures are governed by rare, highly localized collision events. Peak pressures occurred as short-lived concentrated patches rather than distributed loading, and the burst-like resistance and pressure records reflected transitions among sliding/bypassing, overturning, and intermittent local jamming, with the strongest bursts linked to transient load-bearing contact networks and force-chain-like transmission in compacted floe clusters.
4. Both resistance and collision pressure exhibited low-level-dominated responses with persistent heavy tails, indicating that extremes are controlled by rare events. Exceedance analysis showed that the probability of high instantaneous resistance increased rapidly with ice concentration and the tail lengthened markedly at high η. Event-based separation further indicated that collision-induced pressure samples occupied only a small fraction of the record yet dominated the upper tail; increasing θ and η made the collision-pressure tail more persistent by promoting frontal compaction/jamming, while increasing towing speed mainly amplified the intensity of extreme pressure events.

CRediT authorship contribution statement

Ziwang Chen: Writing – original draft, Visualization, Validation, Software, Project administration, Investigation, Formal analysis, Data

curation. **Baoyu Ni:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization. **Shaocheng Di:** Visualization, Validation, Software, Methodology. **Yanzhuo Xue:** Resources, Methodology. **Xuchen Ren:** Validation, Data curation. **Bowen Han:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge funding support from Harbin Engineering University, Norwegian University of Science and Technology and the National Natural Science Foundation of China (Nos.52521009, W2541018, 52371270, 52401318, 52192690 and 52192693).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.oceaneng.2026.125460.

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